EE4-14
EE9-SO16

IMPERIAL COLLEGE LONDON

DEPARTMENT OF ELECTRICAL AND ELECTRONIC ENGINEERING
EXAMINATIONS 2016

MSc and EEE/EIE PART IV: MEng and ACGI

Corrected copy

SPEECH PROCESSING

Thursday, 12 May 10:00 am

Time allowed: 3:00 hours

There are FOUR questions on this paper.

Answer ALL questions.

All questions carry equal marks

Any special instructions for invigilators and information for candidates are on page 1.

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1. a) i) For a discrete-time signal $x(n)$, write down the formula for its complex cepstrum $\hat{x}(n)$. [2]

ii) Discuss the main consequences of performing signal processing in the complex cepstral domain and explain why this is potentially attractive in speech signal processing. [3]

b) Consider a system represented by the rational z-transform

$$X(z) = \frac{A\prod_{k=1}^{M_a}(1-a_k z^{-1})\prod_{k=1}^{M_b}(1-b_k^{-1}z^{-1})}{\prod_{k=1}^{N}(1-c_k z^{-1})}$$

in which $|a_k|, |b_k|, |c_k| < 1$.

i) Identify the terms of this expression corresponding to

- zeros inside the unit circle in z,
- zeros outside the unit circle in z,
- poles inside the unit circle in z. [3]

ii) Hence factorize $X(z)$ into 3 factors:

- a delay factor involving $z^{-\lambda}$,
- a maximum phase factor $X_{\max}(z)$
- a minimum phase factor $X_{\min}(z)$

and write expressions for λ, $X_{\max}(z)$ and $X_{\min}(z)$. [3]

iii) Given that $\hat{X}(z)$ is defined as the complex logarithm of $X(z)$, and adopting a suitable definition for the complex logarithm, write an expression for $\hat{X}(z)$ and hence write expressions for $\hat{x}(n)$ for the cases

- $n = 0$
- $n < 0$
- $n > 0$.

You may use the identity $\log(1-\alpha) = -\sum_{n=1}^{\infty}\frac{\alpha^n}{n}$ for $|\alpha| < 1$.

[6]

iv) Find the complex cepstrum $\hat{x}(n)$ of

$$x(n) = a^n u(n)$$

in which $u(n)$ is the unit step function and $|a| < 1$. [3]

2. a) i) Draw a block diagram that illustrates in overview the main components of an automatic speech recognition system. [2]

ii) Explain what is meant by unigram, bigram and trigram language models. For a recognizer with a vocabulary of 30,000 words, state the number of probabilities that must be estimated for unigram, bigram and trigram language models. [3]

iii) Consider a sequence of words denoted w and a speech signal denoted s. The conditional probability of the word sequence given the speech signal is $pr(w|s)$. Express this probability in terms of a language model and an acoustic model of speech production by employing Bayes' theorem. [3]

b) In a particular application, the speech utterances are restricted to contain only the words 'red', 'blue' and 'green'. All utterances contain at least one of these words. Table 1 gives the frequencies $N(i,j)$ of each possible pair of successive words for which word i is followed by word j in a representative sample of utterances, i.e. $N(i,j)$ is the number of times word j follows word i. 'Start' and 'end' represent the start and end of an utterance.

The unigram probability for word j is denoted $p_u(j)$.

Bigram probabilities $p_b(i,j)$ for all relevant i,j are given by

$$p_b(i,j) = \begin{cases} \dfrac{N(i,j) - d(i)}{N(i)} & \text{for } N(i,j) > 2 \\ b(i)p_u(j) & \text{for } N(i,j) \leq 2 \end{cases}$$

where $N(i)$ is the number of times word i occurs.

The term $d(i)$ is defined as

$$d(i) = \begin{cases} 0 & \text{if } N(i,j) > 2 \text{ for all valid next words } j \\ 0.5 & \text{if } N(i,j) \leq 2 \text{ for all valid next words } j \end{cases}$$

and the term $b(i)$ is chosen so that

$$\sum_j p_b(i,j) = 1.$$

		Word j			
		'red'	'blue'	'green'	'end'
Word i	'start'	10	10	30	0
	'red'	1	0	60	20
	'blue'	3	50	2	10
	'green'	67	5	10	20

Table 1 Word sequence frequencies

i) Calculate the unigram probabilities for each of the words 'red', 'blue', 'green' and 'end'. [3]

ii) Find the values $d(i)$ for all words i. [2]

iii) Hence find the bigram probabilities $p_b(i, j)$. [5]

iv) Briefly explain why the term $d(i)$ is needed for bigram probability calculation and summarize its effect on the bigram probabilities. [2]

3.

a) Speaker recognition aims to determine the most likely identity of an unknown talker from an example of their speech by comparing *features* of the unknown talker's speech to *features* previously extracted from the speech of a set of talkers with known identity. The identity of the unknown talker is determined as the closest matching known talker.

Describe the processing steps employed to extract appropriate *features* from speech. Discuss the desired properties of such *features*. Give examples of appropriate choices for such *features* for the task of speaker recognition. Include any relevant diagrams to clarify your description. [6]

b) At a particular time, the *features* from part (a) are stored in a column vector $\mathbf{z} = (z_1, z_2, \ldots, z_N)^T$ where the elements of $\mathbf{z}$ are assumed independent identically distributed Gaussian random variables having zero mean and unit variance with a probability density function

$$p(z_i) = (2\pi)^{-1/2} \exp\left(-\frac{z_i^2}{2}\right)$$

and ^{T} indicates matrix transpose.

i) Derive an expression for the probability density function of the vector $\mathbf{z}$ and show that the expected value $E\left(\mathbf{z}\mathbf{z}^T\right)$ is equal to the identity matrix. [4]

ii) Consider a non-singular $N \times N$ matrix $\mathbf{A}$. If $\mathbf{x} = \mathbf{A}\mathbf{z}$, obtain an expression for the covariance matrix $\mathbf{C} = E\left(\mathbf{x}\mathbf{x}^T\right)$. [4]

iii) Derive an expression in terms of $\mathbf{C}$ for the natural log of the probability density function of the vector $\mathbf{x}$, where $\mathbf{x}$ is defined in part (ii) above. [6]

Note: You may assume that if the probability density function of $\mathbf{z}$ is $f(\mathbf{z})$, then the probability density function of $\mathbf{x} = \mathbf{A}\mathbf{z}$ is given by $f\left(\mathbf{A}^{-1}\mathbf{x}\right) \times |\mathbf{A}|^{-1}$.

4. a) Speech enhancement can be performed in the STFT domain using the formula

$$Z(l,k) = H(l,k)Y(l,k)$$

in which l is the time-frame index, k is the frequency index, $Y(l,k)$ is the noisy speech signal and $H(l,k) = \sqrt{1 - \frac{\hat{\phi}_v(l,k)}{|Y(l,k)|^2}}$.

i) State what $\hat{\phi}_v$ represents and explain two alternative methods for finding $\hat{\phi}_v$ in a speech enhancement system. Discuss the relative merits of each approach. [5]

ii) What additional parameter(s) does model-based spectral enhancement depend on and how might it/they be estimated? [2]

iii) The values of $Y(l,k)$ over a particular interval are given in Table 1. Assuming $\hat{\phi}_v = 0.16$ and is constant, on which time-frequency indices does the method fail and why? How can this be overcome? [6]

	$l=1$	$l=2$
$k=1$	0.332-0.009i	-0.277+1.695i
$k=2$	0.668+0.482i	-0.226+0.580i
$k=3$	-0.103-0.342i	0.254+0.359i
$k=4$	2.069+1.284i	-2.912+0.738i
$k=5$	-0.009+1.260i	-0.583-0.120i

Table 1 STFT analysis of noisy speech $Y(l,k)$

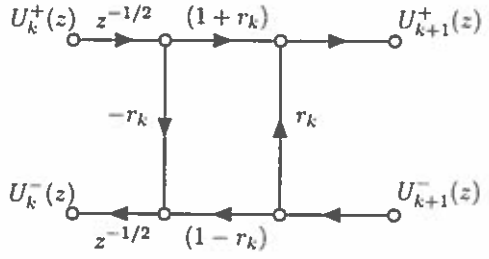


Figure 4.1

b) Consider the lossless tube model of the vocal tract for which a partial sketch diagram is shown in Fig. 4.1.

i) Draw a complete labelled diagram of a lossless tube model of order 4 and describe the model and its relationship to the human speech production system. [4]

ii) For the partial model shown in Fig. 4.1, find expressions for $U_k^+(z)$ and $U_k^-(z)$ in terms of $U_{k+1}^+(z)$ and $U_{k+1}^-(z)$. [3]